Hide menu
Course Instructions
Welcome to the Course: Remote Sensing Data Acquisition, Analysis and Applications
Biography of the Instructor
Text of Slide Audios for Module 1
Solutions to End of Lecture Self Checking Quiz Questions for Module 1
Welcome to Module 1: Acquiring images and understanding how they can be analysed
Week 1 Lectures and Quiz
Week 1 Quiz
Quiz: Week 1 Quiz Submitted

Congratulations! You passed!

Grade received 100%
Latest Submission Grade 100%
To pass 50% or higher
Go to next item

Week 1 Quiz

Submit your assignment
Try again
Due Jun 23, 11:59 PM IST Attempts 3 every 2 hours

Receive grade
To Pass 50% or higher
Your grade 100%
View Feedback
We keep your highest score

1. Scattered sunlight is the most common source of energy used to create images of the earth's surface in remote sensing. True or false? **1 / 1 point**

- false
- true (selected)

Correct
That is correct. Just as we see the landscape through scattered sunlight so do most remote sensing imaging instruments. In the case of thermal and radar imaging scattered sunlight is not used, which we will see later in the course, but the majority of remote sensing applications depend on sunlight..

Like Dislike Report an issue

2. The earth's atmospheric layer: **1 / 1 point**

- scatters sunlight
- selectively absorbs sunlight
- both absorbs and scatters sunlight (selected)

Correct
That is correct. Both absorption and scattering occur

3. The visible and reflected infra red wavelength ranges together are called: **1 / 1 point**

- the reflected wavelengths
- the optical wavelengths (selected)
- the transmitted wavelengths

Correct
That is correct. We will use this term a lot in the course.

4. An atmospheric window is a region of the electromagnetic spectrum in which: **1 / 1 point**

- sufficient reflected sunlight, or energy emitted by the earth itself, reaches the sensor such that images can be formed (selected)
- energy can flow from the earth to the sensor, but not from the sun to the earth
- the atmosphere is totally transparent

Correct
That is correct. Windows don't need to be 100% transparent to be of value for imaging.

5. Satellites are the preferred remote sensing platforms when **1 / 1 point**

- we want cost-effective imaging over large areas (selected)
- we want greatest control over the region being imaged.
- we want to minimise the effect of the atmosphere on the recorded imagery.

Correct
That is correct. Satellites, because of their altitude, view a large region, typically 100km or more in width, and thus can cover the whole earth in a reasonable time frame. Thus they provide data that can be used in many applications in the most cost effective manner.

6. Remote sensing satellites take about 90 minutes to orbit the earth and yet cover the whole earth in about one or two weeks. That is possible because **1 / 1 point**

- the orbital plane is shifted after each completed orbit
- the earth is rotating under the satellite (selected)
- both of the above.

Correct
That is correct.

7. A scan across an image swath recorded by a line scanner takes a finite time, during which the platform is moving. This will show up in the image as **1 / 1 point**

- a distortion in the geometry across the scan line
- a distortion in the geometry in the along-track direction (selected)
- a broadening of the image swath

Correct
That is correct. The region being imaged shifts sideways during acquisition, so that scan lines at the end of the acquisition time will be in their wrong locations laterally compared with those at the start of the scan.

8. In a push broom scanner the images recorded at different wavelengths **1 / 1 point**

- are offset from each other by the IFOV of the detector
- in principle, exactly match each other geometrically (selected)

Correct
That is correct.

9. In a hyperspectral sensor the images at each of the hundreds of different wavelengths are **1 / 1 point**

- recorded sequentially
- recorded simultaneously (selected)

Correct
That is correct.

10. An image is completely described technically by **1 / 1 point**

- IFOV, FOV, frame size, radiometric resolution and number of bands (selected)
- FOV, swath width, radiometric resolution and number of bands
- pixel size, swath width, radiometric resolution and number of bands

Correct
That is correct. The set of defining elements is complete.